Advanced Excel Assignment

Question 1: Explain the difference between Absolute, Relative, and Mixed Cell Referencing in Excel with examples.

Answer:

Cell referencing tells Excel where to look for data. There are three main types:

Reference Type	Description	Example	Behavior when Copied
Relative	Changes based on the relative position of rows and columns.	`A1`	If copied down one row, it becomes `A2`.
Absolute	Remains fixed to a specific cell, no matter where the formula is moved.	`$A$1`	Always stays `$A$1`.
Mixed	Either the row or the column is fixed, while the other remains relative.	`$A1` or `A$1`	`$A1` keeps column A fixed; `A$1` keeps row 1 fixed.

Question 2: **What is a Macro in Excel? How does it help in automation?**

Answer:

A **Macro** is an action or a set of actions that you can run as many times as you want. It is recorded using **VBA (Visual Basic for Applications)** code in the background to automate repetitive tasks.

How it helps in automation:

- **Time-Saving:** Performs complex sequences of formatting, data entry, or calculations in seconds.
- **Consistency:** Eliminates human error by ensuring the exact same steps are followed every time.
- **Efficiency:** Allows users to perform bulk operations (like generating 100 reports) with a single click or keyboard shortcut.

Question 3: What are Text Functions in Excel? Mention any five with examples.

Answer:

Text functions are used to manipulate and clean strings of text. Here are five commonly used ones:

1. **`UPPER`**: Converts all text to uppercase.
   - *Example:* `=UPPER("excel")` → **EXCEL**
2. **`LOWER`**: Converts all text to lowercase.
   - *Example:* `=LOWER("DATA")` → **data**
3. **`PROPER`**: Capitalizes the first letter of each word.
   - *Example:* `=PROPER("john doe")` → **John Doe**
4. **`CONCAT`**: Joins two or more strings together.
   - *Example:* `=CONCAT("Hello ", "World")` → **Hello World**
5. **`LEN`**: Returns the number of characters in a string.
   - *Example:* `=LEN("Excel")` → **5**

Question 4: Scenario Manager in Decision Making

Scenario Manager is a "What-If Analysis" tool in Excel that allows you to create, store, and switch between different sets of input values (scenarios) to see how they affect the results of a formula.

Use in Decision Making:

- **Risk Assessment:** It helps decision-makers evaluate the impact of different levels of risk. For example, a business can compare a **Best Case**, **Worst Case**, and **Most Likely Case** for a project's profitability.
- **Comparison of Strategies:** You can store multiple business strategies (e.g., "High Marketing Spend" vs. "Reduced Production Cost") as different scenarios and quickly toggle between them to see which yields the highest profit.
- **Variable Sensitivity:** It tracks how changes in specific variables (like interest rates, material costs, or sales volume) impact the final outcome, making it easier to identify which factors are most critical to success.
- **Reporting with Summary Tables:** Scenario Manager can generate a **Summary Report** that displays all scenarios side-by-side in a new sheet. This is a powerful tool for presenting options to stakeholders during a meeting.

Example: If you are planning a budget, you could create a "Scenario" where your rent increases by 10% and another where your income increases by 5%. Excel will save both sets of numbers, allowing you to see the final balance for each situation without manually re-typing data.

Question 5 : Define the purpose of VLOOKUP and HLOOKUP. How are they different from XLOOKUP? Which among XLOOKUP and INDEX-MATCH is best while usage?

Answer :

- VLOOKUP searches vertically.
- HLOOKUP searches horizontally.
- XLOOKUP searches both ways, left/right, exact match by default.
- INDEX+MATCH is flexible and fast for older Excel.
- Best for modern usage: XLOOKUP.

Question 6 : Create a dataset of 8 employees with joining dates. Use the dummy dataset to calculate their experience in years and months.

To calculate the experience in years and months, we use the `DATEDIF` function. This function calculates the difference between two dates based on a specific unit ("Y" for years, "YM" for months excluding years).

Answer:

Employee Dataset and Calculations

[Employee Dataset and Calculations link]

Question 7 : You are provided with a dataset containing details of 100 road accidents.Design a Road Accident Analysis Dashboard using Pivot Tables and Slicers in Excel.

Answer:

[Road_Accident_Dataset (Link)]

Question 8 : Create a table of 10 products with stock levels. Use Macros to highlight: ● Low stock (less than 10 units) in red ● Overstock (more than 50 units) in green.

Answer:

[Inventory Stock Link]

Question 9 : You are given a dataset : Create a drop-down list of product names in a separate cell using Data Validation, and write a formula in the adjacent cell so that when a product is selected from the drop-down, its corresponding price is automatically displayed.

Answer:

🛒 **Nova Retail — Product Price Lookup Link**

Question 10 : You are a data analyst working for a retail firm. You've received a sales dataset that includes customer transactions from various regions and product categories. Your job is to analyze this data, clean inconsistencies, extract insights, and build visual reports. Task :

- ☐ **1. Apply data validation in the Units Sold column to ensure only values between 1 and 20 are allowed.**
- ☐ **2. Highlight rows where Profit > ₹5000**
- ☐ **3. Highlight any phone number that is not exactly 10 digits**
- ☐ **4. Find the Revenue generated by each product and also label according to revenue generated as "High", "Medium", "Low".**
- ☐ **5. Find the profit percentage up to two decimal places.**
- ☐ **6. Show Total Revenue and Total Profit by Region.**
- ☐ **7. Show Average Units Sold per Product Category**

Answer:

Sales_Analysis_Report Link